Input system:
Dependent variables:$g_2, p_2, p_3, g_3, g_1, p_1$
Independent variables:t
Equations:

$$\frac{dg_1}{dt} = p_1$$

$$\frac{dg_2}{dt} = p_2$$

$$\frac{dg_3}{dt} = p_3$$

$$\frac{dp_1}{dt} = (-1)k^2(m_0 + m_1)g_1\text{isqrt}(g_1^2 + g_2^2 + g_3^2)^3$$

$$\frac{dp_2}{dt} = (-1)k^2(m_0 + m_1)g_2\text{isqrt}(g_1^2 + g_2^2 + g_3^2)^3$$

$$\frac{dp_3}{dt} = (-1)k^2(m_0 + m_1)g_3\text{isqrt}(g_1^2 + g_2^2 + g_3^2)^3$$

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Result:
Dependent variables:$g_1, g_2, p_2, p_1, p_3, g_3, s_1$
Independent variables:t
Additional variables:

$$s_1 = \text{isqrt}(g_1^2 + g_2^2 + g_3^2)$$

Equations:

$$\frac{dg_1}{dt} = p_1$$

$$\frac{dg_2}{dt} = p_2$$

$$\frac{dg_3}{dt} = p_3$$

$$\frac{dp_1}{dt} = (-1)s_1^3 g_1 k^2 m_0 + (-1)s_1^3 g_1 k^2 m_1$$

$$\frac{dp_2}{dt} = (-1)s_1^3 g_2 k^2 m_0 + (-1)s_1^3 g_2 k^2 m_1$$

$$\frac{dp_3}{dt} = (-1)s_1^3 g_3 k^2 m_0 + (-1)s_1^3 g_3 k^2 m_1$$

$$\frac{ds_1}{dt} = (-1)s_1^3 g_1 p_1 + (-1)s_1^3 g_2 p_2 + (-1)s_1^3 g_3 p_3$$